

Sylhet Engineering College

Department of Civil Engineering

CE 800 (Project/Thesis)

**Undergraduate Research Proposal on
Comparative Analysis of Local Laboratory and Imported Low-Cost
Bitumen for Bangladesh Road Construction**



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Details of the Undergraduate Research Project

Proposed Study Title: Comparative Analysis of Local Laboratory and Imported Low-Cost Bitumen for Bangladesh Road Construction

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Abstract

This study compares laboratory bitumen from Eastern Refineries with imported low-cost bitumen used in Bangladesh road construction, evaluating cost-effectiveness, thickness needs, structural integrity, climate resistance, and load-bearing capacity. Laboratory bitumen shows superior quality and durability, while imported options offer short-term savings but risk premature pavement failure. Findings aim to guide sustainable infrastructure decisions amid Bangladesh's growing highway demands.

Introduction

Bangladesh's road network expansion relies heavily on bitumen for flexible pavements, yet quality variations between local and imported sources affect long-term performance. Imported low-cost bitumen, often from Iran, faces bans by agencies like the Roads and Highways Department (RHD) due to rapid deterioration, while Eastern Refinery's product maintains stable pricing at Tk6,300 per unit. This proposal addresses the need for empirical comparison to optimize costs without compromising safety.

Research Gap

Existing studies focus on general bitumen properties or waste modifiers but lack direct, localized comparisons of Eastern Refinery laboratory-grade versus imported cheap bitumen under Bangladesh's subtropical climate. No comprehensive analysis integrates economic factors with performance metrics like load-bearing in high-traffic areas. This gap hinders policy-making for cost-effective, durable roads.

Significance of the Study

Optimizing bitumen selection reduces maintenance costs and enhances road economy in Bangladesh, where infrastructure supports rapid urbanization. Superior local bitumen improves pavement lifespan, minimizing disruptions and environmental impact from frequent repairs. Results will inform RHD specifications and promote self-reliance in materials.

Research Objective and Questions

Objective: To assess and compare the performance and cost-effectiveness of local laboratory bitumen versus imported low-cost bitumen for Bangladesh road applications.

- What differences exist in penetration, ductility, viscosity, and softening points?
- How do they affect thickness requirements and load-bearing capacity in local climate?
- Which offers better structural integrity and overall economy?

Research Hypothesis

Local laboratory bitumen from Eastern Refineries outperforms imported low-cost bitumen in durability and long-term cost-effectiveness for Bangladesh highways.

Literature Review

RHD standards mandate tests like penetration (60/70 grade preferred), softening point, and viscosity for bitumen quality control. Studies show imported bitumen leads to early rutting, while refinery products ensure stability. Comparative analyses in similar climates confirm higher Marshall stability for premium grades.

Methodology

Collect samples: 60/70 grade laboratory bitumen from Eastern Refinery and equivalent imported bitumen. Conduct RHD-standard lab tests (penetration, ductility, viscosity, softening point, flash point) on triplicate sets. Simulate pavement performance via Marshall stability, flow, and indirect tensile strength tests; analyze cost via lifecycle modeling. Use statistical tools (ANOVA) for comparisons.

Tentative Timeline

- Weeks 1-2: Literature review and sample procurement.
 - Weeks 3-5: Laboratory testing and data analysis.
 - Weeks 6-7: Report writing and dissemination.
- Total: 7 weeks.

Budget

Item	Description	Cost (BDT)
Samples	5 kg local + 5 kg imported bitumen	2000 tk
Lab Tests	Penetration, Marshall, etc. (RHD lab fees)	Free
Equipment	Rental for viscosity tester, etc.	Free
Personnel	Researchers (3 months)	Not Required
Misc	Travel, analysis software	1000 tk
Total		3000 tk

Ethical Considerations

Ensure tests follow RHD safety protocols; dispose samples responsibly to avoid environmental harm. Maintain data integrity without bias toward local suppliers; disclose funding sources if any.

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